University of Cape Town



Machine Learning Assignment 1

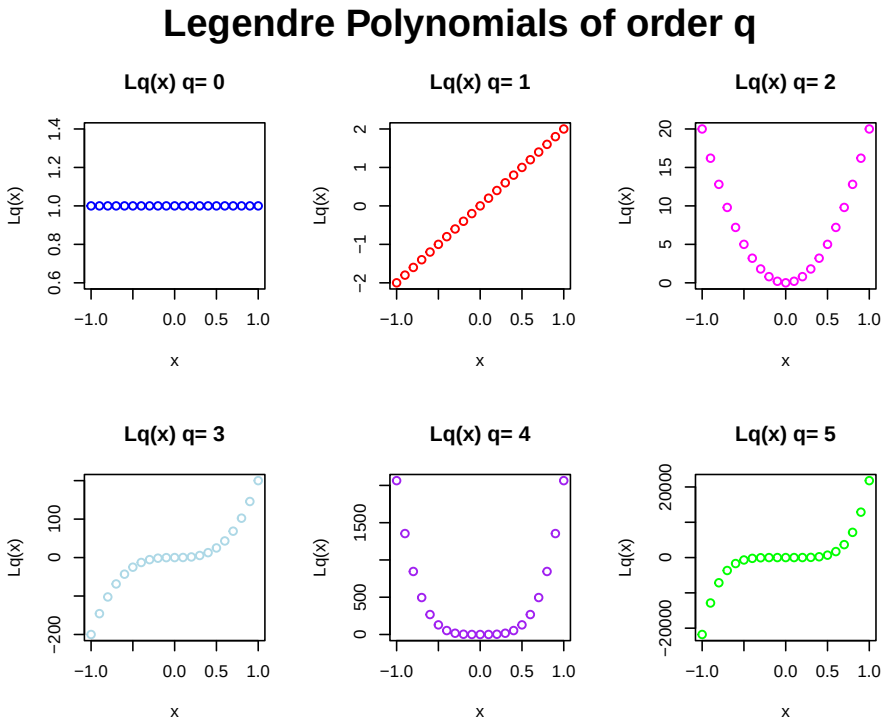
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Question 1

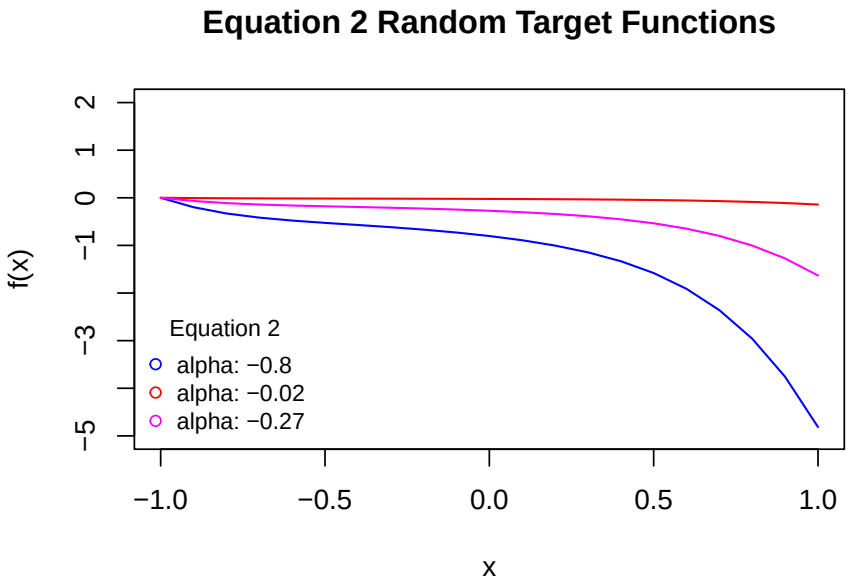
a) Legendre Polynomial Plots

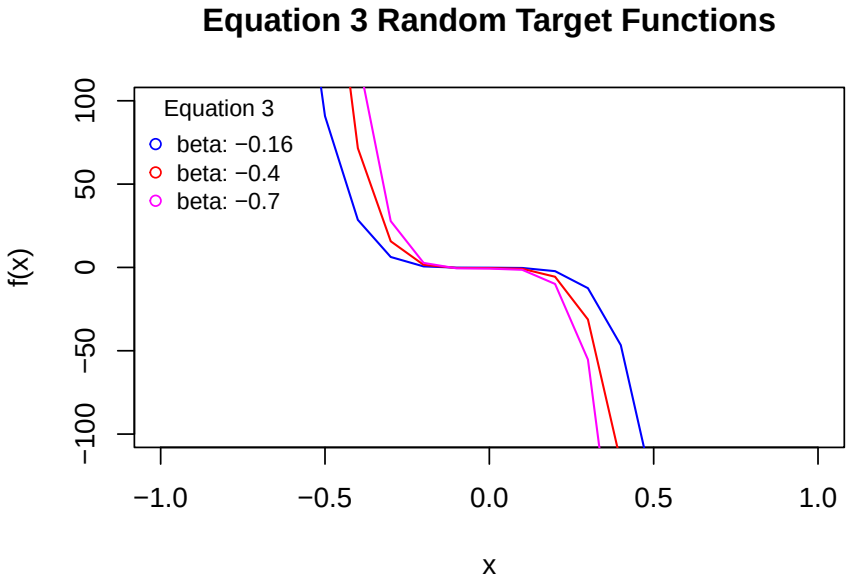


As the order of the polynomial, q, increases, the more nonlinear/complex the$L_q(x)$ gets, albeit slowly.

1. At q=0, $L_q(x)$ devolves to a straight horizontal line $L_q(x)$ =0 over the domain [-1,1].
2. At q=1 $L_q(x)$ starts to have some slope attached, a straight line of the form $y = 2x$
3. At q=2, 4 the non-linear curvature/complexity of $L_q(x)$ starts to become more visible particularly towards the edges of the domain [-1,1; function approximates a parabola/quadratic function
4. At q=3,5 non-linear curvature/complexity of $L_q(x)$ both take the same shape, albeit with different magnitudes of $L_q(x)$ over the same domain.

b) Random Target Functions



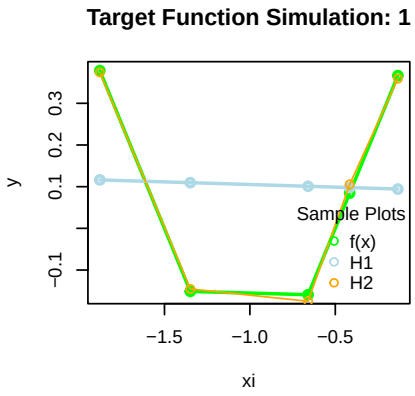


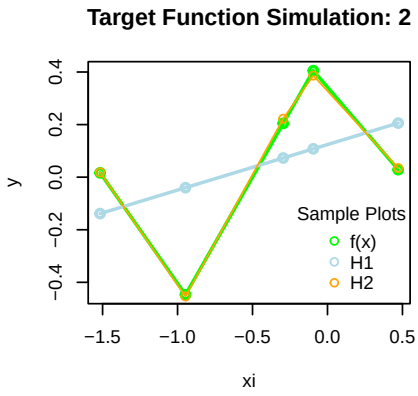
Observations:

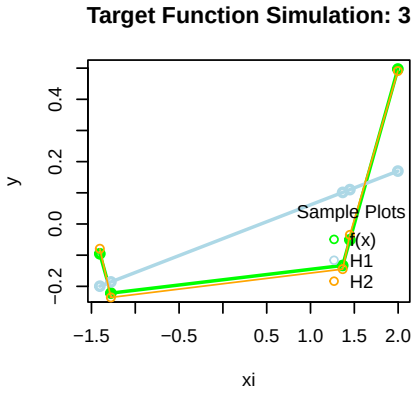
1. For $f(x) = \sum_{q=0}^{Q_f} \alpha_q x^q$ the functions, looks to only ouput negative values over the domian [-1,1]
2. $f(x) = \sum_{q=0}^{Q_f} \beta_q L_q(x)$ outputs both positive and negative values *come-back

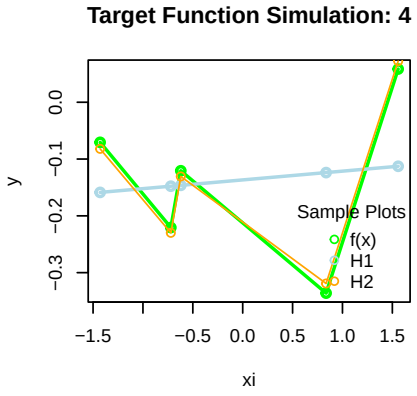
Question 3

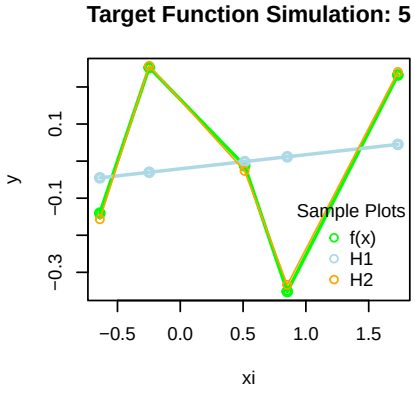
a) Hypothesis Function Simulations









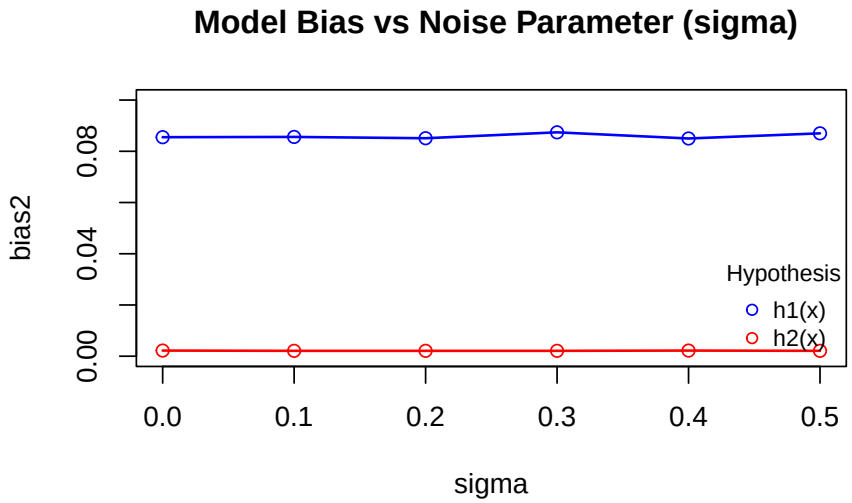


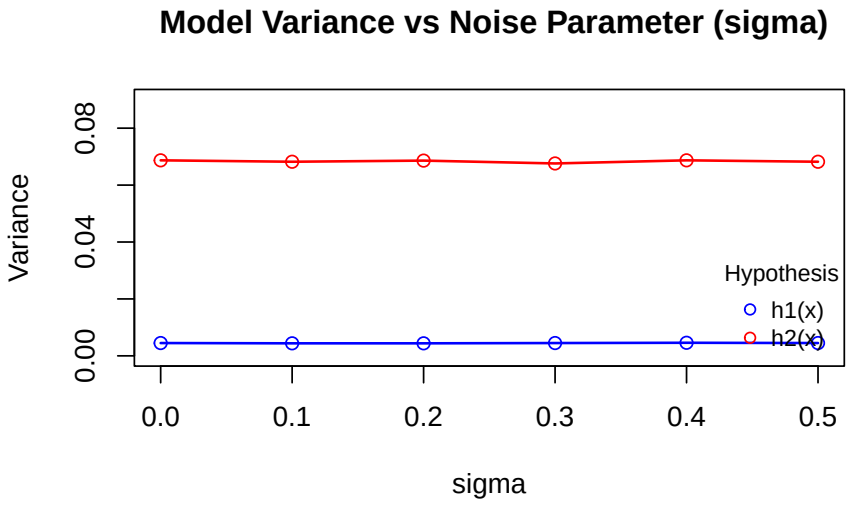
b) A priori expectation

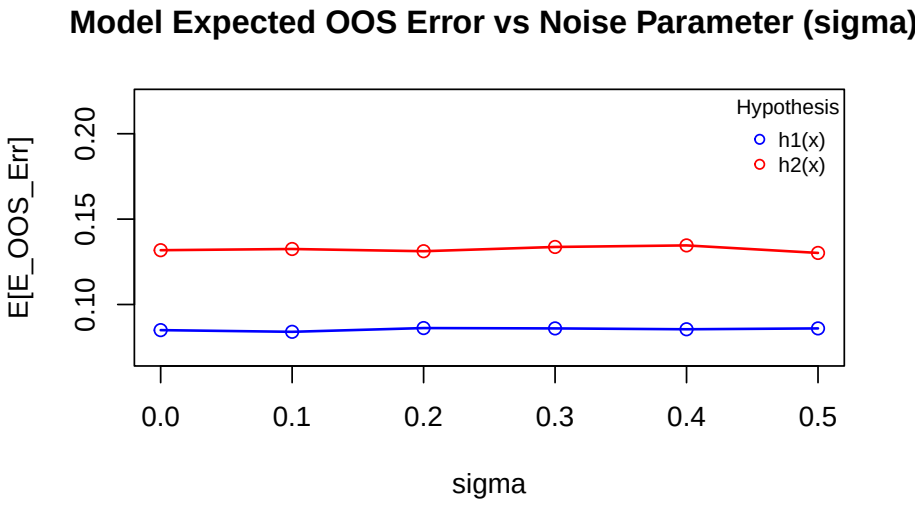
Given that the underlying data generating process is non-linear, the non-linear hypothesis would be more suited to fit the data's non-linear behavior and overall shape.

c) Bias Variance for H1 and H2

	Bias	Variance	Both:Bias+Var	OOS Error	Estimate Differance
h1(x)	0.0854	0.0044	0.0898	0.0841	0.0057
h2(x)	0.0022	0.0703	0.0725	0.1296	0.0571







e) Interpretation *comeback

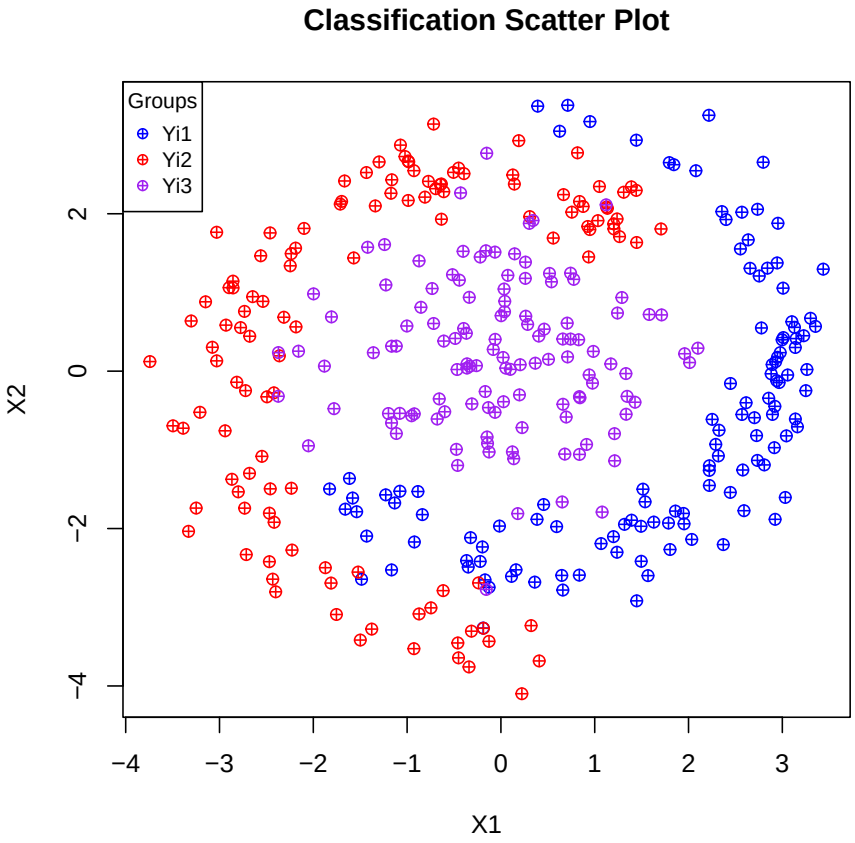
- The expected out of sample error as it relates to the bias and variance of a model is given by

$$E[Err_{oos}] = Bias^2 + Variance + IrreducibleNoise(\varepsilon)$$

- RHS:
  - The linear model , $h1(x)$, has better out-of-sample performance (lower averge OOS error) overall averaged across 1000 simulations over varying values of the noise parameter compared to the more complex model, $h2(x)$.
  -
- LHS:
  - This is also accompanied by a lower average variance across the various noise parameters, this is expected given the that the model is simple relative to $h2(x)$
  - Given the estimated OOS Error, go with the linear model as it is more parsimonious

Question 4

a) Data Set Plot



Observation: The data does not appear to be linearly separable (in the given 2D feature space) as the data does not seem to have linear boundaries between categories/classes. Instead the different set of responses look spiralled around each other. A non-linear neural network seems well suited to capture the non-linear pattern/structure embedded in the data in for the given feature space.

b)

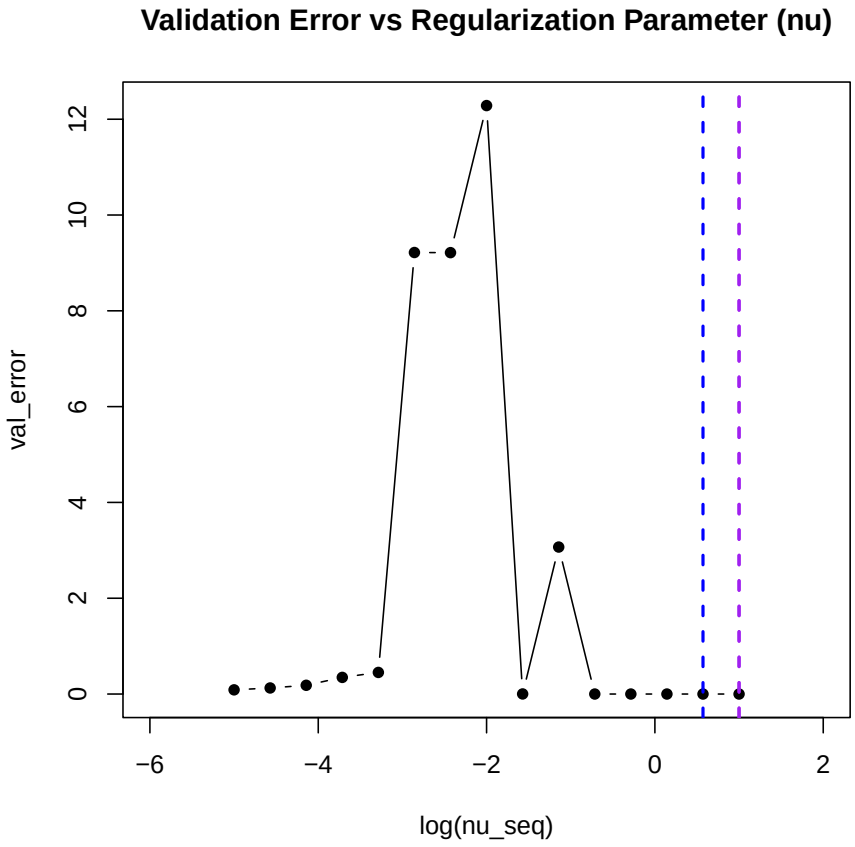
c) Feed Forward Neural Network

Numerical Pitfalls: The computaion of log(yhat) in the objective function may cause the network to explode/ diverge given the undefined behavior of the

log function at 0 (or values very close to 0), which can hamper the numerical stability of the network's computations.

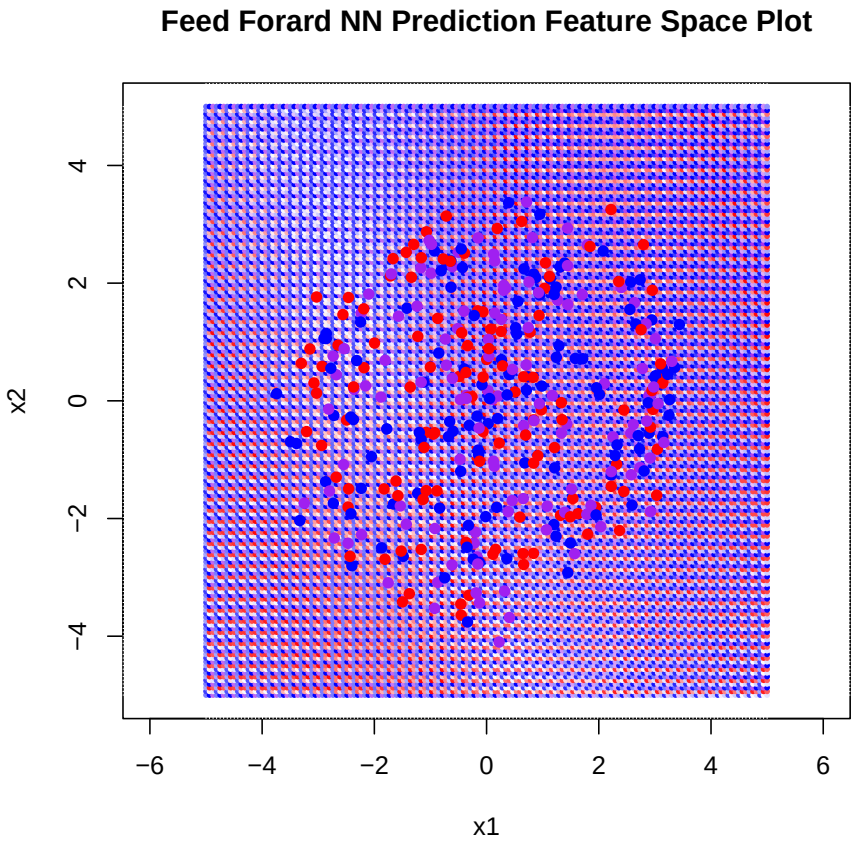
d) Validation Analysis

Plot the above



*comeback Lowest Validation error is obtained at nu=0.007. The closets nu value to the right of the obtained leads to an increase in the validation error, whilst the next best nu (in terms of validation error), produces and more complex model. Thus in this case the lowest nu stays!

d) Response Curve (?)


Feed Forard NN Prediction Feature Space Plot
x2
4
2
0
−2
−4
−6
−4
−2
0
2
4
6
x1